

# Supplementary Material

## Walk-unbiased doubly robust graph propagation for recommendation from exposure-biased logs

Additional results referred to in the main text. Theorem, table and section numbers without the prefix S refer to the main text.

Table S1: Stress tests outside Assumption 1 and with misspecified propensities (log-normal noise with  $\sigma = 0.5$ ). Relative  $|\text{bias}|$  on unexposed candidates.

| Scenario ( $K = 3$ , rel. $ \text{bias} $ on unexposed candidates) | IPS    | IPS + WC | DR     | <b>DRUP</b> |
|--------------------------------------------------------------------|--------|----------|--------|-------------|
| independent exposures (Assumption 1)                               | 0.029  | 0.029    | 1.137  | 0.019       |
| fixed-size slates per user (Proposition 4)                         | 0.219  | 0.219    | 1.119  | 0.024       |
| misspecified propensities, exact imputation                        | 0.568  | 0.568    | 0.000  | 0.000       |
| misspecified propensities, noisy imputation                        | 0.568  | 0.568    | 1.916  | 0.158       |
| independent exposures (Assumption 1), $K = 5$                      | 4.667  | 0.059    | 5.053  | 0.040       |
| fixed-size slates per user (Proposition 4), $K = 5$                | 2.392  | 0.416    | 4.633  | 0.032       |
| misspecified propensities, noisy imputation, $K = 5$               | 19.266 | 1.162    | 16.027 | 0.204       |

Table S2: Tightness of Theorem 6 in a 12-user, 15-item model (20,000 replications, all propensities  $\geq \tau$ ), over all  $(u, i)$ .

|                                                            | $\tau = 0.3$ | $\tau = 0.1$ |
|------------------------------------------------------------|--------------|--------------|
| variance bound (b) / Monte-Carlo variance, median          | 1582         | 8422         |
| variance bound (b) / Monte-Carlo variance, minimum         | 471          | 2126         |
| McDiarmid half-width (d) / empirical 95% quantile, median  | 131          | 1831         |
| McDiarmid half-width (d) / empirical 95% quantile, minimum | 73           | 685          |

Table S3: Exposure-capped re-ranking for all caps. Cells: nDCG@ $K$  (Gini@ $K$ ).

| Coat          | none         | $c = 20$     | $c = 10$     | $c = 5$      | $c = 3$      | $c = 2$      | $c = 1.5$    |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| logged graph  | 0.502 (0.34) | 0.494 (0.34) | 0.501 (0.34) | 0.501 (0.34) | 0.493 (0.33) | 0.494 (0.32) | 0.487 (0.28) |
| IPS adjacency | 0.507 (0.35) | 0.497 (0.35) | 0.506 (0.35) | 0.503 (0.34) | 0.504 (0.36) | 0.493 (0.33) | 0.486 (0.29) |
| DR adjacency  | 0.489 (0.60) | 0.489 (0.60) | 0.489 (0.60) | 0.489 (0.60) | 0.485 (0.58) | 0.475 (0.48) | 0.444 (0.37) |
| DRUP          | 0.489 (0.60) | 0.489 (0.60) | 0.489 (0.60) | 0.489 (0.60) | 0.485 (0.58) | 0.476 (0.48) | 0.443 (0.37) |

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| KuaiRec       | none         | $c = 20$     | $c = 10$     | $c = 5$      | $c = 3$      | $c = 2$      | $c = 1.5$    |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| logged graph  | 0.603 (0.99) | 0.300 (0.91) | 0.228 (0.84) | 0.173 (0.72) | 0.139 (0.60) | 0.115 (0.44) | 0.102 (0.31) |
| IPS adjacency | 0.612 (0.99) | 0.307 (0.94) | 0.234 (0.89) | 0.178 (0.79) | 0.146 (0.66) | 0.121 (0.49) | 0.107 (0.34) |
| DR adjacency  | 0.639 (0.99) | 0.335 (0.95) | 0.257 (0.90) | 0.199 (0.80) | 0.165 (0.67) | 0.140 (0.50) | 0.125 (0.35) |
| DRUP          | 0.638 (0.99) | 0.335 (0.95) | 0.256 (0.90) | 0.199 (0.80) | 0.164 (0.67) | 0.140 (0.50) | 0.125 (0.35) |

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| Yahoo! R3     | none         | $c = 20$     | $c = 10$     | $c = 5$      | $c = 3$      | $c = 2$      | $c = 1.5$    |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| logged graph  | 0.658 (0.28) | 0.659 (0.28) | 0.660 (0.28) | 0.658 (0.28) | 0.656 (0.28) | 0.658 (0.27) | 0.652 (0.23) |
| IPS adjacency | 0.656 (0.27) | 0.657 (0.27) | 0.657 (0.27) | 0.656 (0.27) | 0.656 (0.27) | 0.656 (0.27) | 0.651 (0.22) |
| DR adjacency  | 0.656 (0.27) | 0.657 (0.27) | 0.657 (0.27) | 0.656 (0.27) | 0.656 (0.27) | 0.656 (0.27) | 0.651 (0.22) |
| DRUP          | 0.656 (0.27) | 0.657 (0.27) | 0.657 (0.27) | 0.656 (0.27) | 0.656 (0.27) | 0.656 (0.27) | 0.651 (0.22) |

Table S4: Best nDCG over each operator’s hyper-parameter grid under a cap on the Gini coefficient of top- $K$  exposure.

| Data      | Operator      | Gini $\leq 0.3$ | $\leq 0.4$ | $\leq 0.5$ | $\leq 0.6$ | no cap |
|-----------|---------------|-----------------|------------|------------|------------|--------|
| Coat      | logged graph  | –               | 0.492      | 0.492      | 0.492      | 0.492  |
|           | IPS adjacency | –               | 0.495      | 0.495      | 0.495      | 0.495  |
|           | DR adjacency  | –               | 0.417      | 0.462      | 0.486      | 0.498  |
|           | DRUP          | –               | 0.417      | 0.441      | 0.484      | 0.497  |
| Yahoo! R3 | logged graph  | 0.659           | 0.659      | 0.659      | 0.659      | 0.659  |
|           | IPS adjacency | 0.657           | 0.657      | 0.657      | 0.657      | 0.657  |
|           | DR adjacency  | 0.657           | 0.657      | 0.657      | 0.657      | 0.657  |
|           | DRUP          | 0.657           | 0.657      | 0.657      | 0.657      | 0.657  |

Table S5: Certificates under re-fitted nuisances: the nuisances are re-fitted on the poisoned log (no cross-fitting) and the realised top- $K$  is checked against the frozen-nuisance certificates of the clean system.

| Data      | Operator | fake users $F$ | hit rate (frozen / re-fitted)                      | certified pairs | violated |
|-----------|----------|----------------|----------------------------------------------------|-----------------|----------|
| Coat      | IPS      | 1, 2, 5, 10    | 0.001/0.001, 0.018/0.001, 0.268/0.001, 0.536/0.001 | 217             | 0        |
| Coat      | DR       | 1, 2, 5, 10    | 0.000/0.000, 0.000/0.000, 0.000/0.000, 0.004/0.000 | 5193            | 0        |
| Coat      | DRUP     | 1, 2, 5, 10    | 0.000/0.000, 0.000/0.000, 0.000/0.000, 0.004/0.000 | 5196            | 0        |
| Yahoo! R3 | IPS      | 1, 5, 20, 100  | 0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064 | 0               | 0        |
| Yahoo! R3 | DR       | 1, 5, 20, 100  | 0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064 | 0               | 0        |
| Yahoo! R3 | DRUP     | 1, 5, 20, 100  | 0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064 | 0               | 0        |
| KuaiRec   | IPS      | 5, 50, 200     | 0.001/0.001, 0.001/0.001, 0.133/0.001              | 1616            | 0        |
| KuaiRec   | DR       | 5, 50, 200     | 0.000/0.000, 0.000/0.000, 0.500/0.000              | 2819            | 0        |
| KuaiRec   | DRUP     | 5, 50, 200     | 0.000/0.000, 0.000/0.000, 0.500/0.000              | 2821            | 0        |

Table S6: Jointly private DRUP for all fixed denoising ranks ( $\delta = 10^{-5}$ ).

| Data      | rank                                                                        | $\epsilon = 0.5$ | 1     | 2     | 4     | 8     | 16    | $\infty$ |
|-----------|-----------------------------------------------------------------------------|------------------|-------|-------|-------|-------|-------|----------|
| Coat      | 8                                                                           | 0.397            | 0.403 | 0.430 | 0.441 | 0.455 | 0.459 | 0.452    |
|           | 32                                                                          | 0.391            | 0.402 | 0.416 | 0.434 | 0.454 | 0.457 | 0.454    |
|           | 128                                                                         | 0.370            | 0.390 | 0.412 | 0.440 | 0.436 | 0.452 | 0.453    |
|           | full                                                                        | 0.380            | 0.398 | 0.409 | 0.435 | 0.445 | 0.457 | 0.455    |
|           | non-private DRUP (all nuisances) 0.555; popularity 0.527; 29 public users   |                  |       |       |       |       |       |          |
| Yahoo! R3 | 8                                                                           | 0.387            | 0.445 | 0.491 | 0.522 | 0.558 | 0.581 | 0.597    |
|           | 32                                                                          | 0.397            | 0.437 | 0.480 | 0.518 | 0.551 | 0.574 | 0.608    |
|           | 128                                                                         | 0.397            | 0.433 | 0.472 | 0.505 | 0.535 | 0.561 | 0.610    |
|           | full                                                                        | 0.399            | 0.428 | 0.470 | 0.496 | 0.526 | 0.553 | 0.608    |
|           | non-private DRUP (all nuisances) 0.652; popularity 0.525; 1540 public users |                  |       |       |       |       |       |          |
| KuaiRec   | 8                                                                           | 0.055            | 0.060 | 0.071 | 0.185 | 0.541 | 0.598 | 0.635    |
|           | 32                                                                          | 0.049            | 0.052 | 0.055 | 0.163 | 0.530 | 0.596 | 0.635    |
|           | 128                                                                         | 0.049            | 0.049 | 0.048 | 0.133 | 0.484 | 0.583 | 0.634    |
|           | full                                                                        | 0.050            | 0.054 | 0.061 | 0.089 | 0.163 | 0.316 | 0.635    |
|           | non-private DRUP (all nuisances) 0.637; popularity 0.577; 717 public users  |                  |       |       |       |       |       |          |